Real Time Energy Pipeline Early Warning and Monitoring System

Designed and Developed by:

Rio Sibuea | Alejandro Alvarado | Ifeanyi Njoku

University of Calgary | Master of Data Science and Analytics | Big Data Systems | March 2026

Alberta's Energy Sector: Critical Infrastructure & Its Challenges

~80%	~27%	130K+	1,996
of Canada's oil produced in Alberta	of Alberta's GDP from energy sector	jobs linked to oil & gas	CER-recorded Pipeline incidents

Thousands of km of pipelines transport energy across Canada & the U.S.

The Problem – Pipeline Safety Challenges

- Pipeline failures cause environmental harm, safety hazards and economic losses.
- Traditional threshold-based systems react AFTER anomalies develop.
- Remote Alberta terrain makes manual inspection costly and infrequent.
- No early warning mechanism to alert teams before incidents escalate.

Consequences

- Environmental harm
- Property & injury damage
- Financial & reputational losses
- Regional economic impact

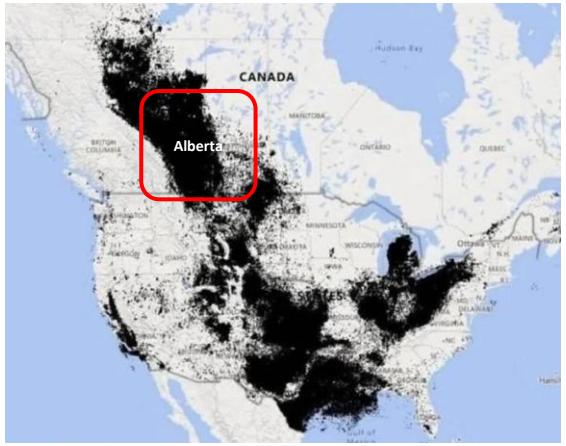


Figure 1. Source: Reddit – *Oil Reserves in States & Canada* map
https://preview.redd.it/oil-reserves-in-states-canada-v0-653bbb4xqlhf1.jpeg

Figure 2. Source: *AI-generated depiction of a pipeline rupture and environmental response.* Generated by Gemini, March 11, 2026.

Source:
Canada Energy Regulator (CER) — Comprehensive Incident Dataset, 2024

Our Solution & Expected Outcome

Proposed Solution

A system to provide **early warning** to the Operation/Security Team to understand potential risks **before** something bad happens to pipelines.

What We Detect

- **Sensor Anomalies** 📊
  - Pressure, Temperature, Flowrate deviations
- **Operating Conditions** 🌡️
  - Out-of-range values triggering ML model
- **Instant Alerts** 📧
  - SNS email/SMS to Operations Team

Expected Outcome

- Highest safety level achieved.
- Financial & reputation loss can be avoided.
- Environmental danger can be mitigated.
- Human injury can be minimized.

Key Value:

- Transform reactive monitoring into proactive, data-driven risk intelligence to protect the people, environment and infrastructure.

Overall Data Engineering end-to-end Pipeline

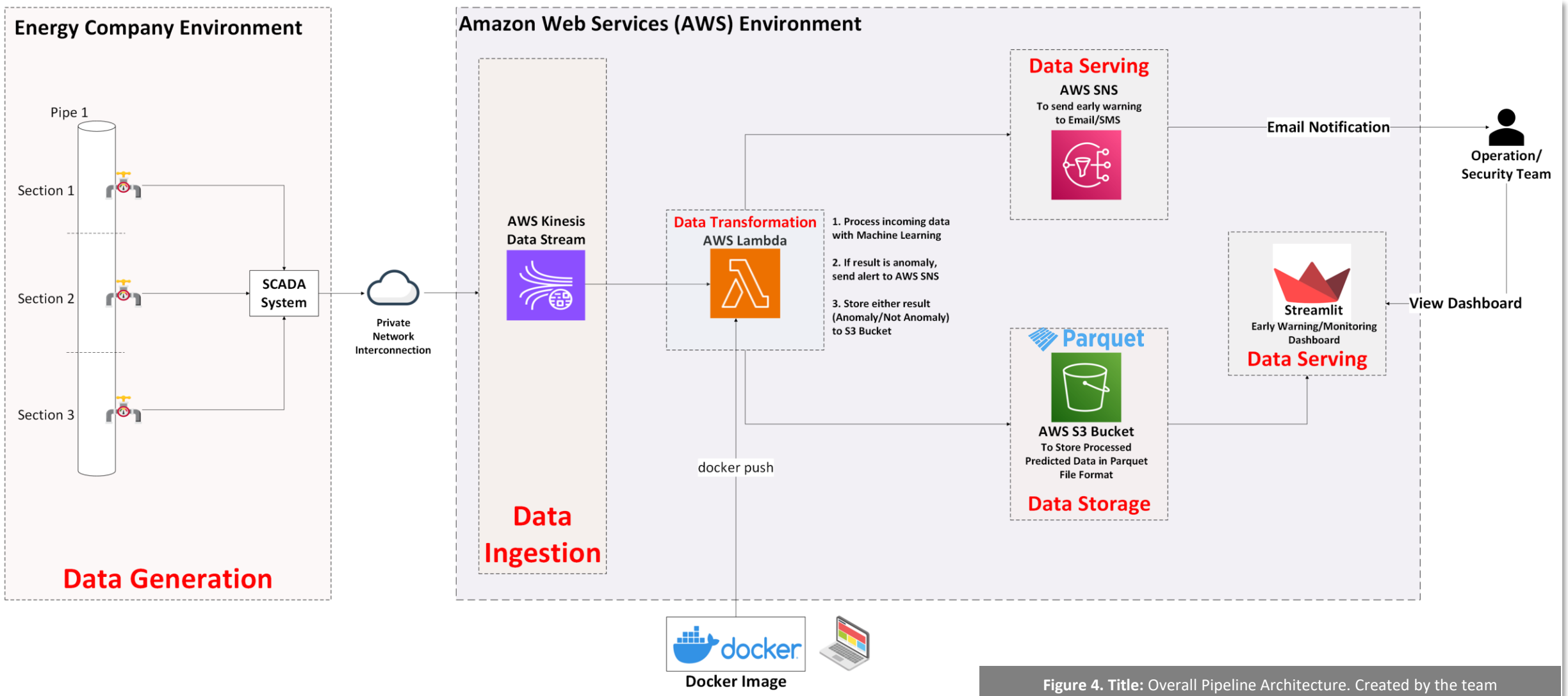
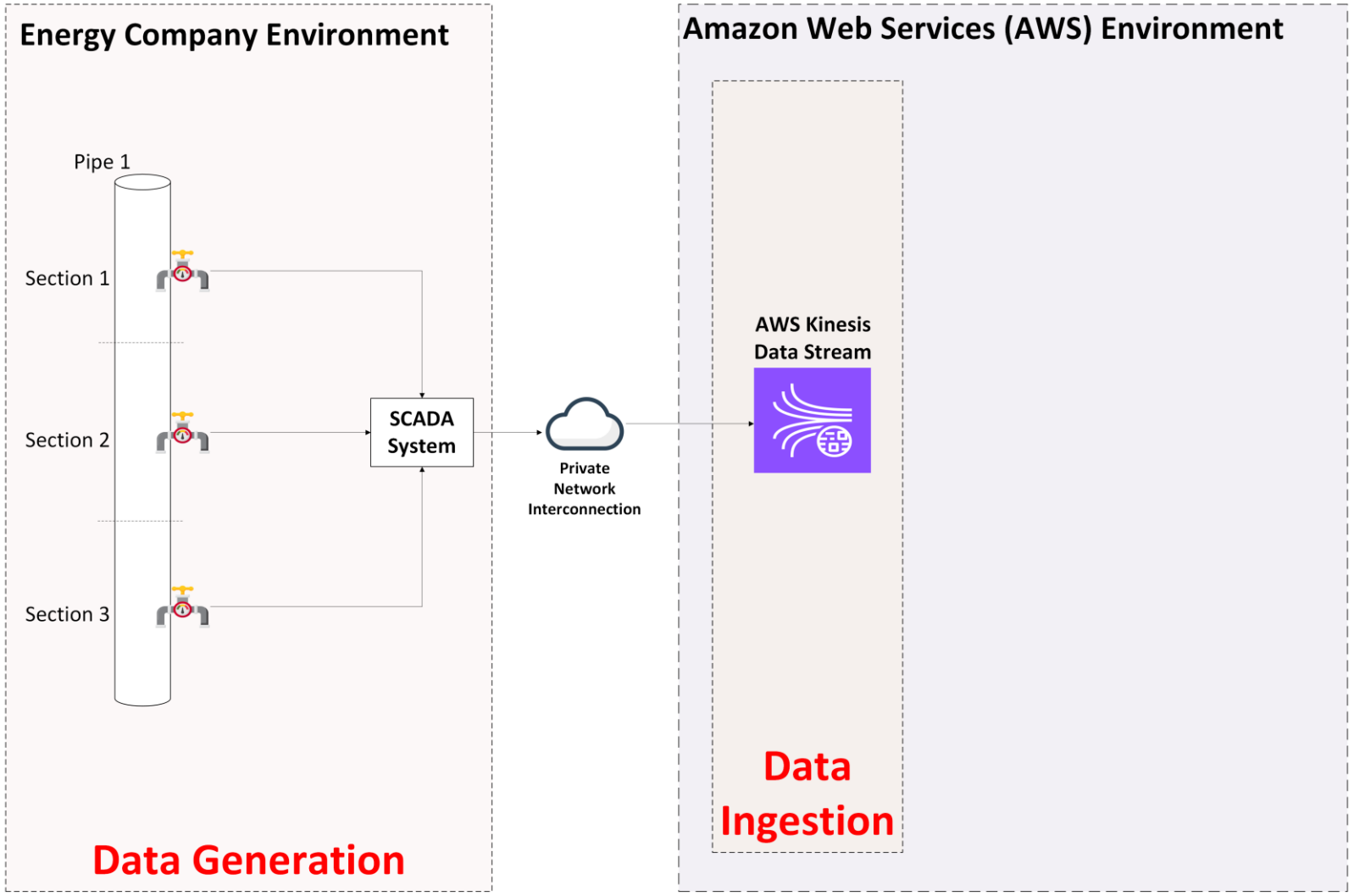
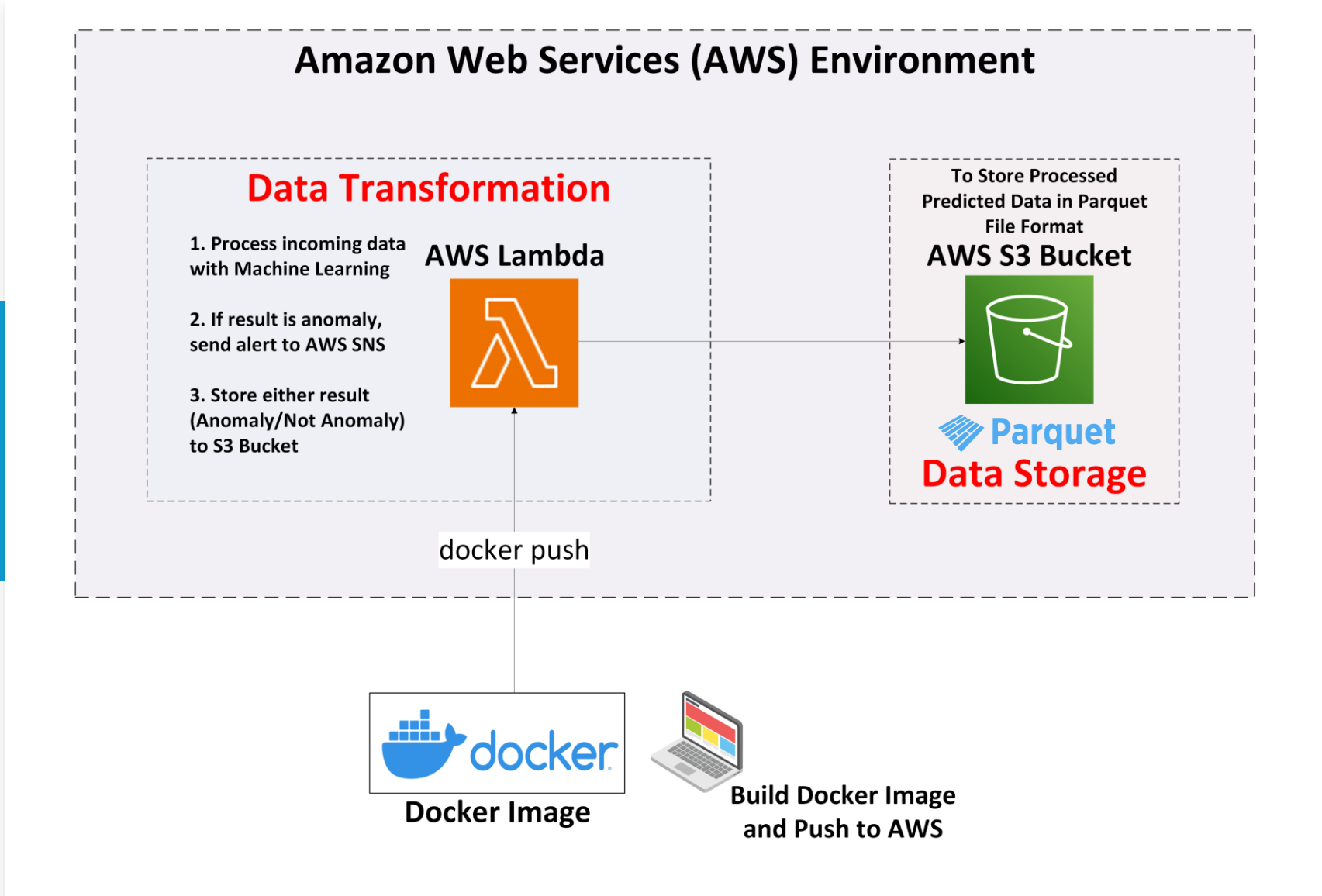


Figure 4. Title: Overall Pipeline Architecture. Created by the team



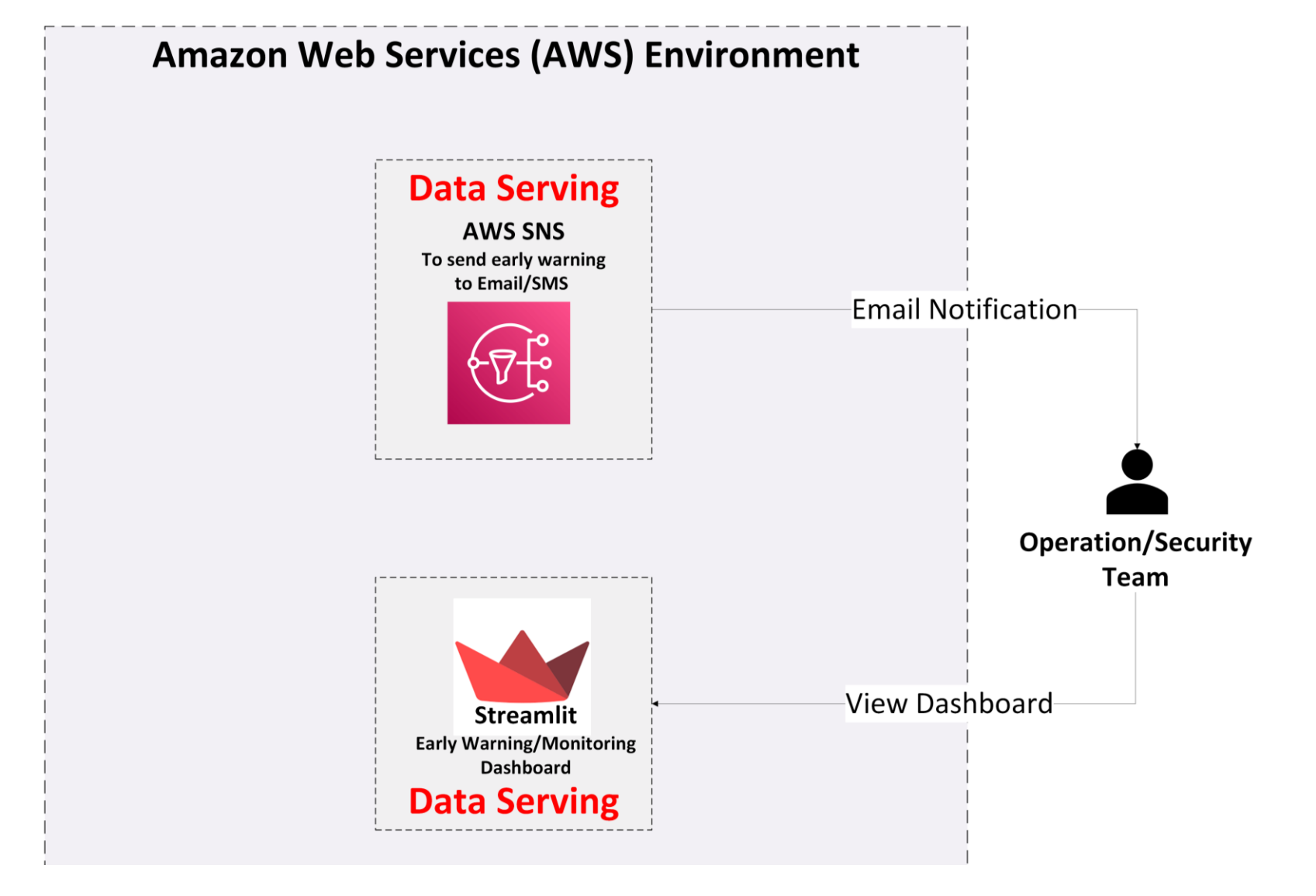
Data Generation and Ingestion

Figure 5. Title: Data Generation & Ingestion. Created by Group 1



Data Transformation and Storage

Figure 6. Title: Data Transformation & Storage. Created by Group 1



Data Serving

Figure 7. Title: Data Serving. Created by Group 1

Success Criteria

✓ End-to-End Data Flow

- Data can flow through the pipelines, be stored in the storage and presented in the Dashboard.

✓ ML Anomaly Detection

- The machine learning model can notify of an anomaly when there is a potential risk.

✓ Email/SMS Alert Delivery

- Anomaly notification is delivered to recipients (email).

A successful project demonstrates working real-time data pipeline + automated anomaly detection + operational monitoring dashboard

Figure 8. Title: *Engineers celebrating on an oil rig at sunset.* **Source:** Image generated by Gemini, March 11, 2026.

Demo

1. Data from the sensor is sent by the simulator and flows to AWS Kinesis.

2. AWS Lambda receives data for processing.

3. **Expectation:**
   - If an anomaly/Critical is detected, an email will be automatically sent to the Operations/Security Team.

   - Dashboard refreshes and visualizes all the processed data, including new status cards and charts.

4. **Value**: **Highest Safety with Early Warning Information –** Operations/Security team alerted before incidents escalate.

Future Work - Roadmap

🎥 Unstructured Data - Video

- CCTV/Drone footage analysis via computer vision.
- Detect intruders (vandalism), wildlife near pipelines.
- YOLOv8 object detection for real-time threat identification.

🛰️ Unstructured Data - Video

- Forest fire detection from satellite imagery.
- Extreme cold and flooding event detection.
- Environmental hazard early warning from remote sensing.

📡 Real Sensor Data Integration

- Replace the simulator with actual SCADA system data.
- CER and Kaggle datasets for model fine-tuning.
- Real-time sensor feeds from oil & gas infrastructure.

🌊 Multi – Purpose Support

- Extend to gas and water pipeline implementations.
- Multi-region AWS deployment for national coverage.
- Scalable architecture for thousands of sensor endpoints.

Figure 9. Title: *On shore oil rig in a morning sunrise.* Source: Image generated by Gemini, March 28, 2026.

THANK YOU FOR YOUR TIME!

Rio Sibuea	Alejandro Alvarado	Ifeanyi Njoku

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GitHub: github.com/cnero101/AWS-Data-Engineering-Project

CER Data: open.canada.ca | Kaggle: kaggle.com/datasets/muhammadwaqas023 | AWS Docs: docs.aws.amazon.com

Figure 10. Title: *On shore oil rig team is having a successful meeting.* Source: Image generated by Gemini, March 28, 2026.